



TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 1 – Project Introduction

Table of Contents

I. Record of Changes.....	3
II. Project Introduction.....	4
1. Overview.....	4
1.1 Project Information.....	4
1.2 Project Team.....	4
2. Product Background.....	4
3. Existing Systems.....	6
3.1 Reference System.....	6
3.2 Existing System.....	15
4. Business Opportunity.....	20
5. Software Product Vision.....	21
6. Project Scope & Limitations.....	22
6.1 Major Features.....	22
6.2 Limitations & Exclusions.....	22
7. Future work.....	22

I. Record of Changes

Date	A* M, D	In charge	Change Description
06/05/2024	A	Nguyen Duc Anh	Create report item(1,2,3,4,5,6)
26/06/2024	A	Trinh Thanh My	Create report item(7)

*A - Added M - Modified D - Deleted

II. Project Introduction

1. Overview

1.1 Project Information

- Project name: A system to improve the quality of customer care services for polyclinics
- Project code: MedPal
- Group name: SEP490-G30
- Software type: Website

1.2 Project Team

Full Name	Role	Email	Mobile
Trinh Tien Dat	Lecturer	dattt67@fe.edu.vn	0346131398
Nguyen Duc Phuong	Member	phuongndhe153051@fpt.edu.vn	0345322806
Nguyen Duc Anh	Leader	anhndhe160131@fpt.edu.vn	0986526526
Nguyen An Thuyen	Member	thuyennahe160369@fpt.edu.vn	0328815402
Ngo Anh Duy	Member	duynahe160133@fpt.edu.vn	0937488102
Trinh Thanh My	Member	myttthe153136@fpt.edu.vn	0327264235

2. Product Background

Currently, the need for health care is a top priority for many people of all ages. During the reform period, young people have also paid more attention to their own health. Therefore, clinics are also improving the quality of their customer care services to meet the increasing needs of customers. Customer service at polyclinics include many different aspects, from welcoming patients hospitality and providing accurate and timely information to providing post-treatment support and follow-up care. Building a friendly and professional health care environment not only helps patients feel secure and satisfied. Especially in the current era, when information and communication technology is developing rapidly, the integration of modern technology in management and customer care also plays a significant role. Online appointment booking systems, remote consultations, and personal health management applications are changing the way clinics interact with and serve their patients.

68A polyclinic , our customer , a polyclinic which uses a traditional approach to manage their clinic. The clinic's staff has 30 people including doctors from different departments, receptionists, security guards and nurses. Doctors in the clinic are usually doctors from large hospitals who come to work in shifts.

A polyclinic is a private clinic that includes at least two specialties: internal medicine and surgery. Besides, the polyclinic may have other specialties such as: emergency resuscitation, dermatology, eye, ENT, etc. In particular, talking about the Department of Internal Medicine, there will be major internal medicine specialties and small internal medicine specialties. First, major internal medicine specialties will be related to large

body parts such as gallbladder, musculoskeletal, neurology,... As for the internal medicine department, there will be specialties such as: ophthalmology, paediatrics, dermatology,...As for the surgical department, there will be surgical departments such as: Department of General Surgery, Department of Neurosurgery, Department of Thoracic Surgery, Department of Gastrointestinal Surgery, Department of Kidney and Urology Surgery,.....

The organisational structure of each department will consist of two key roles. One is the primary doctor, who will work directly with patients throughout their medical examination and treatment process. The other is the support staff, who will assist the doctor during patient interactions and meticulously record all patient information.

Given the size of the clinic, the clinic's medical examination process will be for the patient to come and meet with the receptionist, then state the condition the patient is experiencing and make specific medical examination requests according to the patient's needs with the clinic. The patient will then be assigned a clinical examination and after the clinical examination results are available, the patient will be transferred to a specific department and examined directly with the doctor of that department. The doctor will base on the patient's clinical examination results to make an accurate conclusion about the patient's current medical condition. From there, the doctor will explain the patient's current condition, prescribe medication, explain the prescription in detail and record notes about the patient's condition.

Before implementing the current software, Clinic 68A faced numerous disadvantages that led to various difficulties and errors in its operations. Managing patient data, appointments, and medical records manually on paper easily resulted in mistakes and loss of information, making it difficult to search and retrieve when needed. This increased the risk of medical errors due to the lack of systems to support monitoring and reminders for medication dosages, follow-up appointments, and other medical instructions. Additionally, inefficient appointment management and patient coordination led to long waiting times or overlapping schedules, wasting time and resources. The clinic also struggled with financial and billing management when done manually, leading to errors, financial losses, and difficulties in tracking expenses. The ability to monitor operational efficiency was also limited due to the lack of analytical and evaluation tools, as well as the absence of statistical data and reports to support management decisions. Complying with regulations and medical standards was challenging without a system to maintain and update medical records according to standards, increasing the risk of violating patient privacy and data security. Lastly, communication and interaction with patients were limited without a system for reminders and appointment notifications, leading to patients forgetting appointments or not arriving on time, reducing service quality and patient experience. Implementing management software would help Clinic 68A address these issues, improve operational efficiency, and enhance service quality.

After implementing the current software, Clinic 68A has resolved many issues in data management, finance, and appointment coordination, enhancing operational efficiency and minimising errors. However, the clinic still lacks customer care services, appointment reminders, online consultations, online appointment booking, and the ability for receptionists to create patient accounts. These shortcomings lead to several disadvantages in customer care. The absence of an appointment reminder system causes patients to easily forget important appointments, affecting their treatment

process. The lack of online consultation services and online appointment booking reduces convenience and flexibility for patients, especially those with busy schedules or who live far from the clinic. The inability for receptionists to create patient accounts for tracking and managing personal records is also a significant limitation, reducing the ability to provide comprehensive and personalised customer care services. These drawbacks result in a suboptimal patient experience, impacting service quality and customer satisfaction. Integrating these services would help Clinic 68A enhance the customer experience and overall service quality.

First of all, the primary deficiency lies in the absence of an online medical consultation and examination platform. Because there will be a group of customers who experience some mild symptoms such as cough, flu, etc., so they will have a subjective mentality and concern when going to medical facilities for treatment. see the doctor. Some people cannot arrange a reasonable time to visit medical facilities. Not only that, in case the patient has to have some minor tests done for certain diseases, the patient can communicate directly with the doctor who examined the patient through the chatbox. Therefore, online medical consultation and treatment is a quite necessary form for clinics and patients.

Another glaring concern is the clinic's sole reliance on a single scheduling method - necessitating patients to physically visit the premises to register their information and secure appointment slots. Therefore, customers will have difficulty when they cannot choose an examination schedule that suits their schedule. For that reason, having an additional method of scheduling online examinations will optimize time and effort for patients and clinics. In addition, when there is a method to manage the doctor's work schedule, it is also easier for patients to proactively arrange and schedule appointments in the doctor's available time frames.

For patients who do not have that kind of time, coming to get test results or medical examination results is also an obstacle for them. Because having to travel to the clinic to get results will take up some of their time. Consequently, the clinic ought to implement an online system for disseminating results, thereby conserving valuable time for all parties involved.

Recognizing the limitations in the clinic's current customer care service. Hence, **MedPal** was conceived to surmount the remaining deficiencies that medical facilities are confronting. It represents an optimal solution geared towards enhancing the quality of customer service delivery.

3. Existing Systems

3.1 Reference System

3.1.1 Med247

Med247 link: <https://med247.vn/>

Description: The website is developed and used by Med247 with branches in Hanoi such as: Med247 Hai Ba Trung: No. 7-9 Thanh Nhan, Hai Ba Trung District,

Med247 Tay Ho: House D5 Dreamland Tay Ho, 107 Xuan La, Bac Tu Liem District,

Med247 Nam Tu Liem: Moc Lan 6-28 Vinhomes Greenbay, Nam Tu Liem District.

Med247 provides patients with a comprehensive medical experience (in person to

online), from a high-quality polyclinic system to remote medical consultation services via Video Call & Chat in a super convenient application. App Med247. Med247's operating model is a convenient chain clinic model. Patients have access to high-quality medical services anytime, anywhere, 24/7.

3.1.1.1 Med247's Main System Actors

A.Sales Representatives (Salesman)

This position is responsible for meeting with customers and providing guidance and direction, using the system to connect customers and doctors. They will organize and participate in meetings with customers to understand their needs and desires regarding healthcare services. Additionally, they will present detailed information about Med247's services, including the appointment booking process and doctor options, and answer related questions.

Guiding customers on how to use the system to book appointments, manage health information, and track treatment schedules is also a crucial part of the job. Furthermore, they will use Med247's management system to create connections between customers and suitable doctors, ensuring that contact information and appointments are updated accurately and promptly.

They will assist customers in the appointment booking process and resolve any issues that arise regarding meeting with doctors. With these responsibilities, they will be a key link in helping customers access healthcare services conveniently and effectively, while enhancing the healthcare experience at Med247.

B.Admin

The role of this position encompasses daily management and supervision responsibilities, ensuring the smooth operation of various administrative and medical tasks.

This includes meticulously managing doctors' schedules to optimise their availability and efficiency, as well as arranging work assignments for individuals to maintain a balanced and productive workflow. Additionally, this role involves overseeing and managing connections between different departments, facilitating effective communication and collaboration to ensure seamless operations across the organisation.

By coordinating these critical tasks, this position plays an essential role in maintaining the overall functionality and harmony within the organisation, thereby contributing to the delivery of high-quality healthcare services.

C.Doctor

The role of the Doctor is integral to the medical team, as they are responsible for examining, diagnosing, and treating patients with a wide range of conditions. These professionals conduct thorough medical examinations and administer treatments, ensuring that each patient receives the best possible care. Additionally, they work

closely with nurses to complete patient information, ensuring accurate and comprehensive medical records. Beyond the immediate treatments, Doctors offer valuable medical advice to patients, guiding them on health management and preventive care. They also oversee and manage patient care, coordinating with other healthcare professionals to ensure a cohesive and effective approach to each patient's health needs. By fulfilling these responsibilities, primary healthcare providers play a crucial role in maintaining the health and well-being of the community

3.1.1.2 Med247's Key Features

A.No-Wait Appointment Booking

The "No-Wait Appointment Booking" feature allows patients to freely choose their doctor and schedule appointments directly through the application on their phone, eliminating the need for traditional appointment booking processes through the clinic. Patients can browse a list of doctors, view detailed information about their specialties, working hours, and reviews from previous patients, and select the doctor that best fits their needs and preferences.

After selecting a doctor, patients can immediately book an appointment for their examination or treatment without waiting or going through multiple intermediaries. This feature not only enhances convenience and flexibility for patients but also reduces waiting times and improves overall satisfaction with the healthcare service.

B.Personalized Health Checks and Recommendations

The "Personalized Healthcare Experience" feature of the system aims to provide a tailored healthcare experience for each patient, offering appropriate medical advice and treatments.

The system analyses individual health data, including medical history, current symptoms, and genetic factors, to deliver the most suitable recommendations and treatment plans. By utilising artificial intelligence and advanced analytical algorithms, the system can propose personalised treatment plans, monitor health progress, and provide necessary reminders.

C.Online Payment

Med247 offers robust support for online payment systems, significantly streamlining financial transactions for patients. This service simplifies the payment process, allowing patients to settle their medical bills conveniently and efficiently from the comfort of their homes.

By integrating online payment options, Med247 reduces the need for physical visits solely for financial transactions, saving time and reducing hassle for patients.

This streamlined process not only enhances patient satisfaction but also ensures quicker processing of payments, benefiting both patients and healthcare providers by fostering a more efficient financial management system.

3.1.1.3 Specific Benefits for Each User Group

A. For Sales Representatives

For sales representatives, the advantages of using Med247 are both significant and transformative. One of the foremost benefits is the continuous and seamless access to vital information. Sales representatives can effortlessly retrieve up-to-date data and insights, which is crucial for making informed decisions and tailoring their sales strategies. Additionally, Med247 provides user-friendly tools that are specifically designed to enhance customer engagement. These tools streamline communication, enable personalised interactions, and facilitate efficient follow-ups, all of which contribute to a more effective sales process. By leveraging these capabilities, sales representatives are empowered to build stronger relationships with clients, address their needs more precisely, and ultimately drive higher sales performance. This integrated approach not only simplifies their daily tasks but also significantly boosts their ability to meet and exceed sales targets.

B. For Admin

For managers, the benefits of using Med247 are considerable. One major advantage is the efficient management of information intake, ensuring that all incoming data is organised and easily accessible.

This leads to a significant improvement in response times, allowing administrators to address issues more quickly and effectively.

Consequently, the overall workflow becomes more streamlined, contributing to a more efficient and responsive working environment.

C. For Doctors

Utilising a patient record management system offers significant advantages for doctors.

Firstly, easy access to medical records saves time by eliminating the need for manual retrieval of patient information. This not only reduces administrative tasks but also allows doctors to focus more on patient care and treatment.

With an effective system for organising and managing information, doctors can easily track medical history, health indicators, and other crucial details, thereby enhancing the quality of care and making more accurate treatment decisions.

3.1.1.4 The disadvantage of Med247

According to Dr. Bao, who used to work at med247, said: "Med247's system is excellent, but their biggest drawback is that they develop the system solely for their own use, without selling or leasing the software. Even if they were willing to sell or lease it, their potential customers are limited to a segment of large polyclinics. Medium and small clinics have little chance of accessing it, and they often do not need to use all the functions of Med247's system. This creates a significant gap in the market".

Therefore, we see an opportunity to enter this niche market by offering a tailored solution that meets the specific needs of medium and small clinics, providing them with the essential functionalities they require at a more accessible price point. This strategy not only fills the market void but also helps smaller clinics enhance their operational efficiency and patient care services.

3.1.1.5 Med247 Workflow

A. Doctors online consulting workflow:

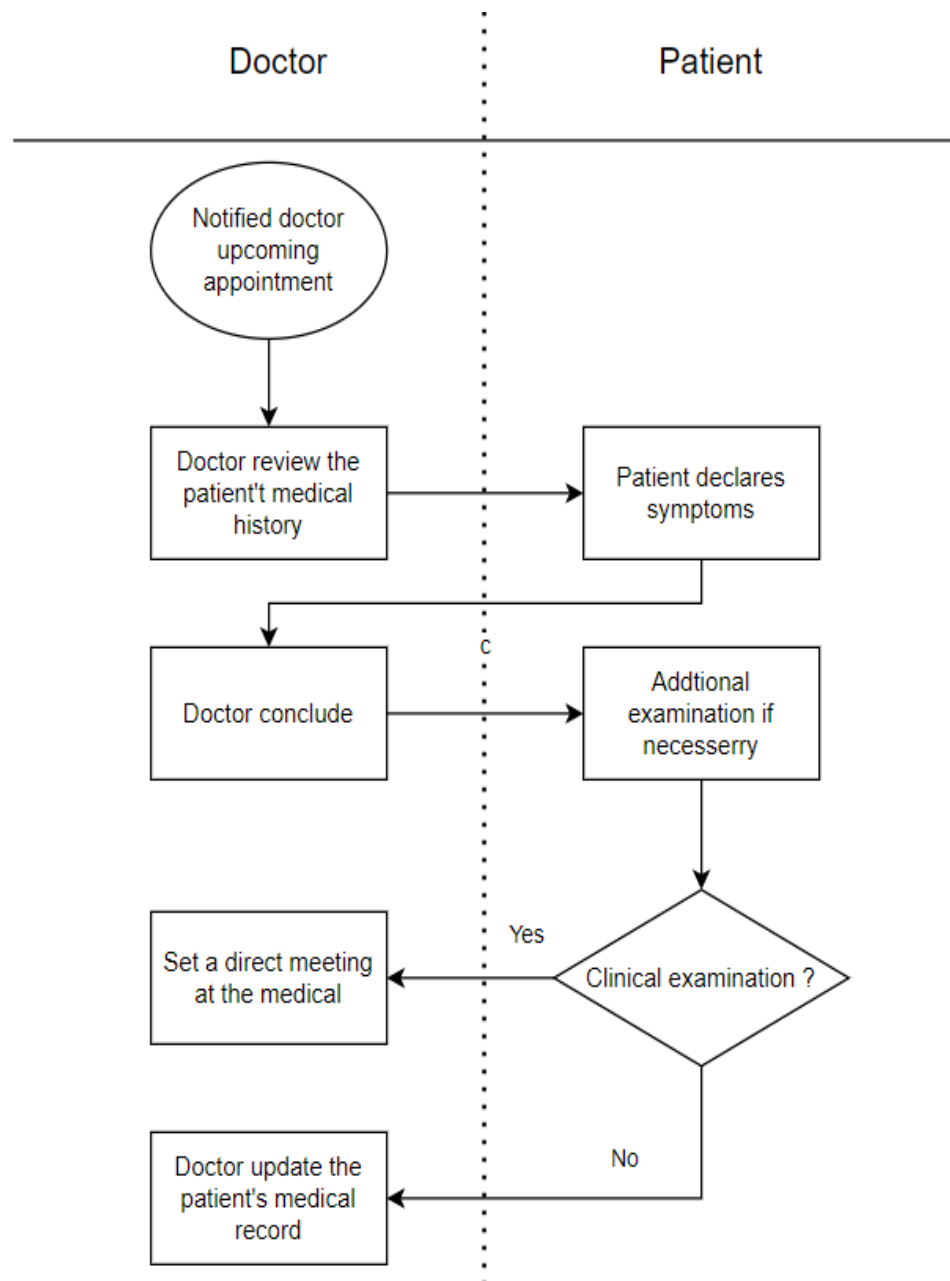


Figure 3.1.1.5.1: Doctor online consulting work flow

B. Patient online consulting workflow:

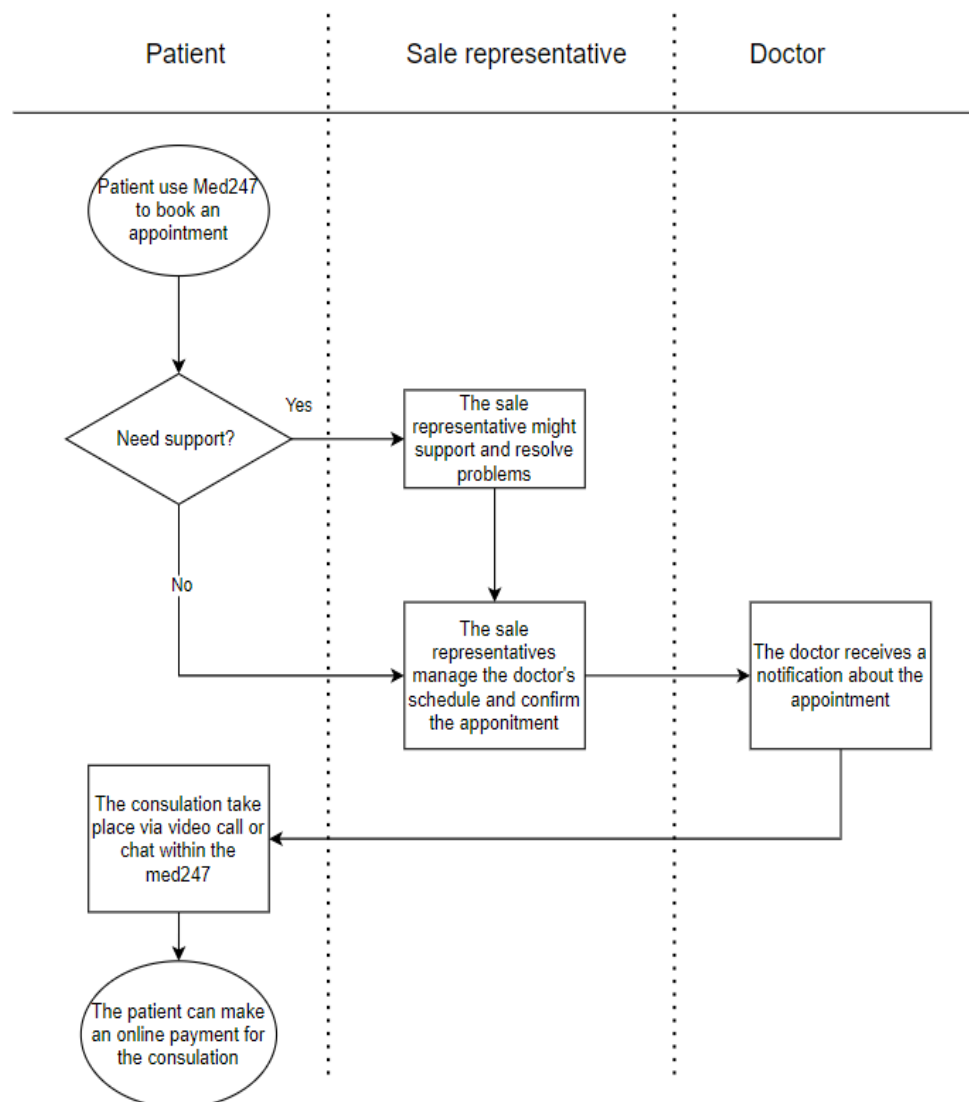


Figure 3.1.1.5.2 Patient online consulting work flow

3.1.2 Medlatec

Medlatec link: <https://medlatec.vn/>

Description: The website is developed and used by Medlatec and they apply this software to 10 branches including their hospitals and clinics in the following areas: Hanoi, Ho Chi Minh City, Vinh Phuc, Thanh Hoa.

Medlatec is a leading healthcare system in Vietnam, offering a full range of services from home sample collection to in-clinic medical examinations and online consultations. Through its advanced technology platform and high-quality medical facilities, Medlatec ensures patients receive top-notch medical care anytime, anywhere.

3.1.2.1 Medlatec's Main System Actors

A. Medical Representatives (MedRep):

Medical Representatives (MedRep) play a crucial role in the Medlatec organisation, serving as the primary liaison between customers and the range of services offered. Their main responsibility is to engage with customers to thoroughly understand their individual needs and requirements.

By doing so, they provide tailored information and guidance about Medlatec's diverse services, ensuring that customers are well-informed and able to make the best decisions for their healthcare needs. MedReps are also instrumental in facilitating seamless connections between customers and healthcare providers, bridging any gaps and ensuring that the process is smooth and efficient.

Their role involves not only providing expert direction and support but also actively listening and responding to customer feedback. This comprehensive approach not only enhances customer satisfaction but also strengthens the overall service delivery, making Medlatec's offerings more accessible and effective for those seeking medical care.

B. Admin Manager:

The Admin Manager position at Medlatec is responsible for overseeing the daily operations of the organisation, including managing the schedules of doctors and nurses to ensure all activities are organised and effective. This role involves arranging and coordinating daily tasks to ensure that all processes and operations are carried out on schedule, while maintaining effective connections and coordination between departments. The role requires excellent organisational skills, time management abilities, and strong communication skills to ensure the smooth and efficient functioning of the healthcare facility.

C. Doctor:

Doctors serve as the cornerstone of healthcare, taking on the pivotal role of primary healthcare providers. Their responsibilities are comprehensive and vital to patient well-being. Each day, doctors conduct thorough medical examinations, utilising their expertise to diagnose and treat a wide range of health conditions.

Their role extends beyond just performing clinical procedures; they are also responsible for coordinating closely with nurses and other healthcare staff to ensure that all patient information is accurately recorded and managed. This collaboration is essential for providing cohesive and effective patient care. In addition to their clinical duties, doctors offer valuable medical advice, guiding patients through their treatment options and helping them make informed decisions about their health.

Their commitment to managing patient care involves not only addressing immediate health concerns but also planning and overseeing long-term treatment strategies. Through their dedicated efforts, doctors play a crucial role in enhancing patient outcomes and maintaining high standards of healthcare.

3.1.2.2 Medlatec's Key Features

A. No-Wait Appointment Booking:

The "No-Wait Appointment Booking" feature on the Medlatec application allows patients to select their preferred doctor and schedule appointments directly without having to go through intermediate steps.

This feature enables users to easily choose a doctor based on specialty, availability, and other personal criteria. The system automatically checks the doctor's availability and provides open time slots for users to select.

This not only enhances the convenience of scheduling appointments but also reduces waiting times and improves the efficiency of healthcare facilities. Instead of waiting through phone calls or other communication channels, patients can complete their appointment booking quickly and efficiently.

B. Personalized Health Checks and Recommendations:

The "Personalized Health Checks and Recommendations" feature on the Medlatec system offers users a completely personalised healthcare experience. Based on each patient's health profile and personal information, the system analyses the data to provide tailored care and treatment recommendations. Considering factors such as medical history, current symptoms, and individual risk factors, the system delivers suggestions and care plans adjusted to each patient's specific needs. This not only ensures that users receive more accurate and relevant medical advice but also enhances treatment effectiveness. Patients will feel that their care is thoughtfully tailored to their health condition, improving their overall quality of life and health.

C. Online Payment:

The "Online Payment" feature of Medlatec makes financial transactions quick and easy. Instead of having to visit a payment counter or use traditional methods, patients can pay for services directly through the Medlatec platform. The system supports various payment methods, from credit cards to bank transfers, ensuring flexibility and convenience for users. With a simple and user-friendly interface, patients can complete their payments in just a few steps, saving time and reducing hassle. This feature not only enhances the patient experience but also streamlines the payment process, making it smoother and more efficient.

3.1.2.3 Specific Benefits for Each User Group

A. For Medical Representatives:

For Medical Representatives, the advantages of utilising Med247 are both substantial and far-reaching. One of the primary benefits is the continuous access to a wealth of information, which is essential for their role. This constant availability of up-to-date data allows Medical Representatives to stay informed about the latest services and offerings, enabling them to provide accurate and relevant information to their clients. Additionally, Med247 equips them with intuitive, easy-to-use tools designed to

streamline and enhance customer engagement. These tools facilitate more effective communication by allowing Medical Representatives to manage interactions efficiently, track customer preferences, and personalise their approach. This streamlined functionality not only simplifies their workflow but also significantly boosts their ability to build strong relationships with clients, ultimately leading to improved service delivery and increased customer satisfaction. By leveraging these capabilities, Medical Representatives can perform their roles with greater efficiency and impact, ensuring that they meet the needs of their clients effectively and professionally.

B.For Admin Managers:

For Admin Managers, the benefits of utilising Med247 are particularly significant and impactful. One of the most notable advantages is the efficient management of information reception. The system is designed to streamline the process of capturing and organising data, ensuring that all incoming information is systematically categorised and readily accessible.

This improved organisation facilitates quicker and more accurate decision-making. Additionally, Med247 enhances response times across various administrative functions. By automating and optimising workflows, the system enables Admin Managers to address inquiries, resolve issues, and implement solutions with greater speed and efficiency. This not only boosts overall productivity but also contributes to a more responsive and agile administrative environment. The ability to manage information effectively and respond promptly to emerging needs ultimately leads to a more streamlined operation, fostering a more efficient and organised workplace that supports the broader goals of the organisation.

C.For Doctors:

For doctors, the benefits of using Med247 are both transformational and invaluable. One of the most significant advantages is the easy access to comprehensive medical records. With Med247, doctors can swiftly retrieve patient information from a centralised, digital repository, eliminating the need for time-consuming searches through physical files. This streamlined access not only saves valuable time but also reduces the likelihood of errors and ensures that doctors have the most up-to-date and accurate information at their fingertips. As a result, doctors can dedicate more of their time and focus to providing high-quality patient care. By minimising the time spent on administrative tasks and information retrieval, Med247 enables doctors to concentrate on diagnosing, treating, and engaging with patients more effectively. This shift enhances the overall efficiency of medical practice, leading to better patient outcomes and a more satisfying experience for both doctors and patients. The system's ability to facilitate seamless access to information ultimately supports doctors in delivering superior care and improving the overall efficiency of their medical practice.

3.1.2.4 The disadvantage of Medlatec

Due to the scale of medlatec including many hospitals and clinics in many areas, the cost of maintaining the website is also a problem that causes difficulties for medlatec

3.2 Existing System

3.2.1 Medical examination process for patients

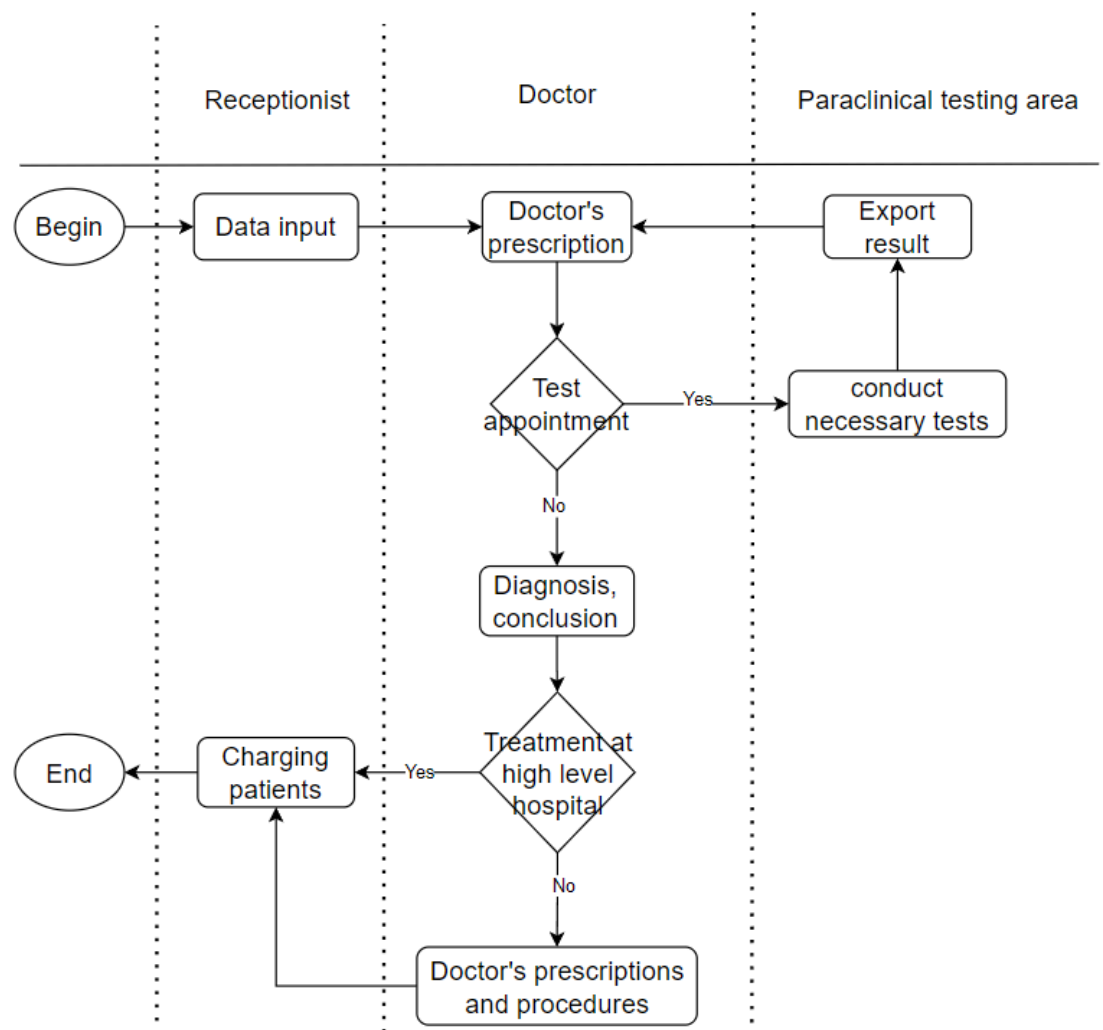


Figure 3.2.1.1 Patient's medical examination and treatment flow chart

Upon arrival for an examination, the patient will be greeted by the receptionist. The receptionist will request the patient's information. For new patients, the receptionist will input and save their details into the system. For returning patients, the receptionist will retrieve their information from the system using their phone number (as illustrated below).

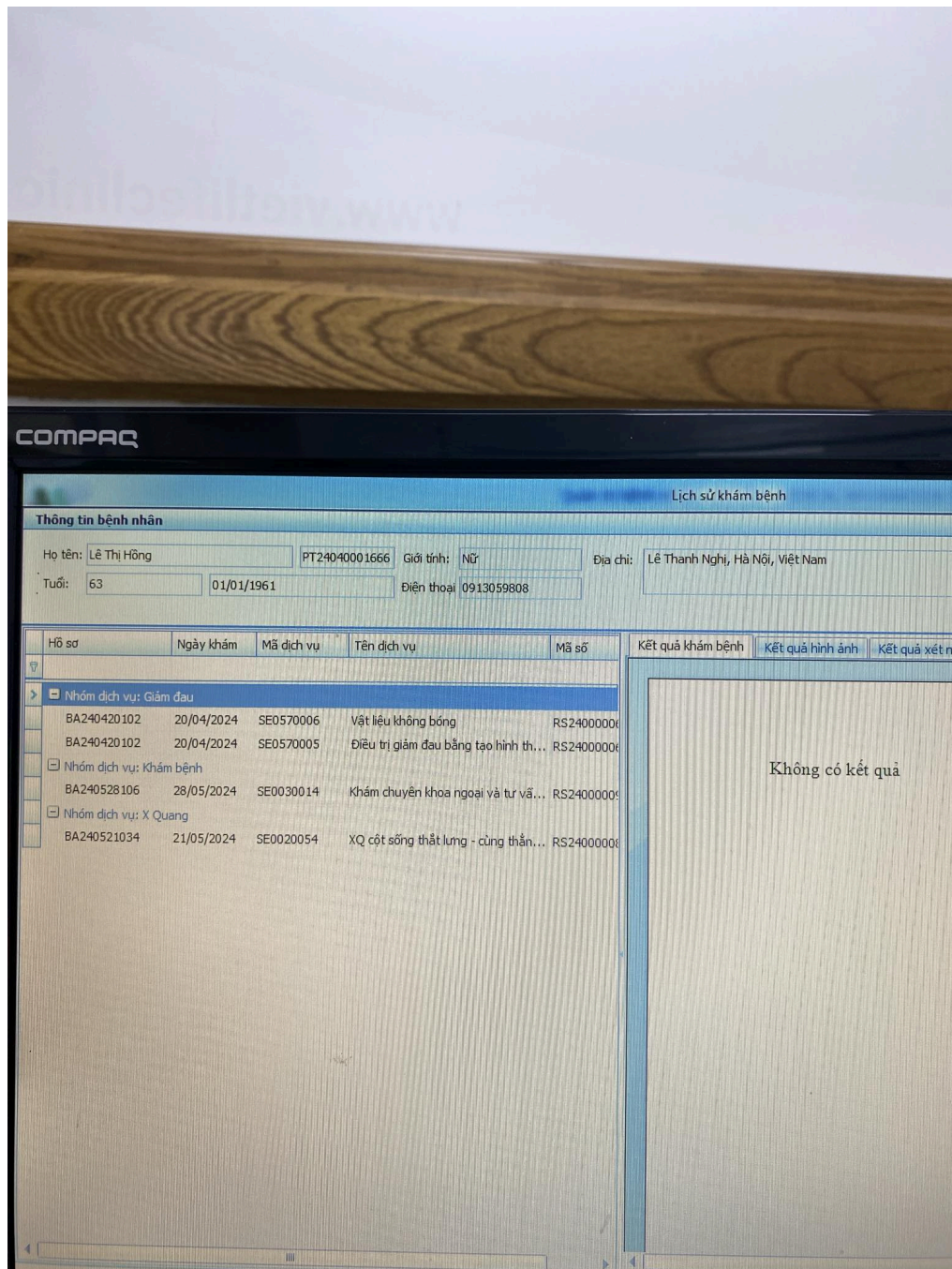


Figure 3.2.1.2 Medical examination history

Quản trị bệnh nhân

Thêm mới

Thông tin chính				Thông tin chi tiết			
Mã bệnh nhân	PT24050002476	Ảnh	Địa chỉ	Thẻ công dân	Thư điện tử		
Tên bệnh nhân	Aa		Quốc tịch	Ngày cấp	Người thân (NT)		
Điện thoại			Tỉnh thành	Nơi cấp	Địa chỉ NT		
Giới tính	Nam		Quận huyện	CHMD	Điện thoại NT		
Ngày sinh	28/05/2024		Xã phường	Nơi cấp	Trình độ		
Tuổi			Dân tộc	Ngày cấp	Nghề nghiệp		
Tiền sử			Ngày đăng ký	Mã số thuế	Nơi làm việc		
Ghi chú				Mạng xã hội	Mặt khẩu		

Lưu Làm mới Thoát Đăng ký hồ sơ Đăng ký dịch vụ

Trang thái nhập tên bệnh nhân							
PT24050002451	Trần Gia Linh	Nữ	22/11/2002	22	Số 4, C04 Giảng Võ	Việt Nam	Hà Nội
PT24050002450	Đinh Quang Tuy	Nam	15/12/1990	74	Đông Hà	Việt Nam	Quảng Bình

1/2144

Người dùng: Nguyễn Thị Hồng Liên - Trang thái: Thêm mới bệnh nhân

Figure 3.2.1.3 Patient profile

The receptionist will use the patient's information to direct them to the appropriate doctor's room. For example, a patient needing dental care will be sent to the dental room, while a patient with stomach or intestinal issues will be directed to the internist's room. The doctor will then diagnose and examine the patient. If further tests or procedures are necessary, the doctor will complete the test request forms and refer the patient to the relevant departments.

PHÒNG KHÁM ĐA KHOA 68A
Số 111 Phòng Hưng - Hà Đông - Hà Nội
ĐT: 024 33 540 112 - 0989.068 259 - 038.22 99 668

PHIẾU XÉT NGHIỆM

Họ và tên: Năm sinh:
Địa chỉ:
Chẩn đoán LS:

☐ Công thức máu -	☐ MD, MC	☐ HBsAg	☐ Nhóm máu (ABO)
☐ Glucose - HbA1c		☐ HCV	☐ T3 T4 TSH FT4
☐ Cholesterol - Tri - HDL - LDL		☐ HIV	☐ HP
☐ GPT - GOT - GGT		☐ HbeAg	☐ Anti HBsAg
☐ Bilirubine TP - TT - GT		☐ ASLO	☐ Định lượng HBV-DNA
☐ Ure - Creatinine - Protein - Abumin	☐ CRP	☐ Nội tiết Hormon	
☐ A.Uric	☐ δ-latex	☐ Dịch niệu đạo	
☐ Amylase	☐ Giang mai	☐ Dịch phụ khoa	
☐ AFP; PSA; CEA; CA125; CA15-3	☐ Ma túy	☐ Soi tinh dịch	
☐ CA19-9; CA72-4; CYFRA21-1	☐ Rubella	☐ Nước tiểu	
☐ SCC; NSE; TG; AntiTG	☐ Dangue	☐ Beta HCG	
☐ Double test -	☐ Triple test	☐ STTB	

☐ Các xét nghiệm khác :.....

Ngày.....tháng.....năm 20.....
Bác sĩ điều trị

Figure 3.2.1.4 Test form

Test results will be exported and given to the patient. They will take the test results back to the original diagnostician's office.

PHÒNG KHÁM ĐA KHOA 68A
Số 111 Phòng Hưng - Hà Đông - Hà Nội
ĐT: 0936.822.886

PHIẾU KHÁM NỘI SOI TAI - MŨI - HỌNG
Số phiếu: 00001123

Tên bệnh nhân: PHUONG Thuần 30 Tuổi: 30 Giới tính: Nữ
Địa chỉ: HÀ NỘI Ngày khám: 11/11/2023
Lý do khám: KT Khám lần thứ: 1 Nữ khoa bệnh:

HÌNH ẢNH NỘI SOI

Tai phải Tai trái Tai phải

MÔ TẢ KẾT QUẢ NỘI SOI
Tai: Vành tai, ống tai tai bình thường; Màng nhĩ hai bên sáng, không sưng.

Chẩn đoán: Hình ảnh nội soi tai bình thường.
Điều trị:
Lời dặn của BS:

Ngày 09 Tháng 11 Năm 2023
Y. Bác sĩ nội soi
Nguyễn Hoàn

* Lưu ý: Mang phiếu khám khi đến khám lần tiếp theo

PHIẾU TRA LỜI KẾT QUẢ
XÉT NGHIỆM MÔ BỆNH HỌC

NGƯỜI XÉT NGHIỆM: NGUYỄN VĂN HỮU
PHÒNG XÉT NGHIỆM GENEHUNG
Số 111 Phòng Hưng - Hà Đông - Hà Nội
ĐT: 0936.822.886

KẾT QUẢ XÉT NGHIỆM

Họ và tên: NGUYỄN VĂN HỮU
Số ID bệnh: 123456789
Số ID xét nghiệm: 123456789
Ngày xét nghiệm: 11/11/2023
Địa chỉ: HÀ NỘI

TÊN XÉT NGHIỆM	KẾT QUẢ	ĐƠN VỊ	YÊU CẦU	CHÚA CHỈ	PHÂN LÃNH
1. Histopathology (H&E)	2.53	mm			
2. Immunohistochemistry (IHC)	1.75	mm			
3. IHC	0.41	mm			

Chẩn đoán: Hình ảnh nội soi tai bình thường.
Điều trị:
Lời dặn của BS:

Ngày 09 Tháng 11 Năm 2023
Y. Bác sĩ nội soi
Nguyễn Hoàn

PHÒNG XÉT NGHIỆM TỔNG HỢP
Số 111 Phùng Hưng - Phúc La - Hà Đông
ĐT 02433.540.112 * 0936.922.886

Họ và tên: Bùi Tam Điệp
Tuổi: 38
Địa chỉ: Hà Nội
Chẩn đoán: KT

Xét nghiệm công thức máu 18 chỉ tiêu

TRUNG TÂM CÔNG DỤNG CÔNG NGHỆ Y HỌC
PHÒNG KHÁM DA KHOA 68A
Số 111 - Phùng Hưng - Phúc La - Hà Đông - Hà Nội
SĐT: 02433.540.112
BỘ: ĐIỆP - SĐT: 0-52
TEST: GLUCOSE

TEST	RESULT	UNITS	EXPECTED VALUE
GLUCOSE	17.7	mmol/L	(3.8 - 6.4)

Bác sĩ chỉ định: TTC
Bác sĩ xét nghiệm: [Signature]
Ngày: 20/10/2024

PHÒNG KHÁM BỆNH DA KHOA 68A
Địa chỉ: Số 111, Đường Phùng Hưng, P. Phúc La, Q. Hà Đông, TP Hà Nội
Điện thoại: 024. 33 540 112 - 0977 851 785 - 0936 922 886
Email: phoongkhamda68a@gmail.com

BỆNH VIỆN 183
Khoa GPB
ĐT: 02433.540.112
Số bệnh lý: 09475

Họ và tên: NGUYỄN THỊ AN
Tuổi: 64
Giới: Nữ

Giới thiệu ngày: 27/10/2024
Địa chỉ: HÀ NỘI
Bệnh phẩm: Tuyến giáp
Chẩn đoán LS: Bướu giáp
Đại thể: Khối bệnh phẩm tuyến giáp KT 3x2 cm, mặt cắt có các nhân nang 0.6-1 cm, màu vàng nhạt

Nhận xét vi thể:
Tuyến giáp gồm các nang tuyến giáp tăng sinh, tăng sinh bào đực, tế bào nang là biểu mô tế bào nhân tròn và đều, lõi đậm có 1 tầng sinh vi dày và 1 tầng đồng sinh thể cholesterol.

Chẩn đoán giải phẫu bệnh lý: Bướu giáp đơn thuần
Ngày 28 tháng 10 năm 2024
Bác sĩ Đọc vi thể: [Signature]
TIS. BS. Nguyễn Thị Linh

Figure 3.2.1.5 Test result form

After receiving the test results, the patient will return to the doctor's room for the initial examination, presenting the results for diagnosis. At this point, there are two possible scenarios: if the patient's condition is too severe for the clinic to manage, the doctor will refer them to a higher level of care. If treatment is feasible at the clinic, the doctor will prescribe medication or recommend a procedure. The patient will then proceed to the front desk to pay for the tests and examination fees. The receptionist will record the payment information in the patient's file and issue an invoice (The clinic will not be dispensing medications. Patients will need to fill their prescriptions at a pharmacy outside of the clinic).

3.2.2 Other Business Flows

A. Doctor's Work Schedule Management:

The clinic manager plays a crucial role in organizing and overseeing the doctors' work schedules to ensure smooth operation and efficient patient care. On the clinic manager's schedule management screen, comprehensive information is displayed, including each doctor's full name, department, and working days and hours. This organized display allows the clinic manager to effectively coordinate the doctors' availability, manage any changes or adjustments, and ensure that patients have access to the care they need. By maintaining a well-structured schedule, the clinic can optimize its resources, reduce waiting times, and enhance the overall patient experience.

B. Financial Management:

Effective financial management is essential for the smooth operation of the clinic. The accountant consolidates all financial information, including patient numbers, billing data, and expenses. This comprehensive data collection ensures that every financial transaction is accurately recorded.

The accountant then compiles detailed financial reports and presents them to the clinic management, providing transparency and a clear overview of the clinic's financial health.

3.2.3 Exploitable gaps

This model remains in use due to its low operating costs, ease of use, and comprehensive management features. However, it lacks several important customer care functionalities:

- First, when there are too many patients coming in for examinations, they have to wait a long time, which impacts the quality of their medical care.
- Second, patients scheduled for follow-up visits might forget their appointments, leading to changes in their treatment plan and potentially worsening their condition.
- Third, there are no online examination services, resulting in the omission of a significant number of potential patients nationwide.
- Fourth, according to a survey of doctors at the clinic, it is a common occurrence for patients to lose their medical records, so they need a place to store and look them up online.
- Fifth, patients needing to monitor their health after a clinic visit must either contact the doctor themselves or return to the clinic. This can be inconvenient, as these methods may not provide a quick response to their needs.

Finally, this software is only used for internal management, not for patients, which can cause a lack of transparency in the medical examination and treatment process..

4. Business Opportunity

4.1 Business Opportunity

Small and medium-sized clinics often rely solely on internal management software for patient care, leading to missed opportunities and reduced satisfaction. Patients experience difficulty booking appointments online, miss important reminders, and lack the ability to see online, impacting revenue and customer experience. Meanwhile, large clinics like Med247 or Medlatec do not lease or sell their websites. Even if they did, their products would not be suitable for small and medium-sized clinics.

MedPal places a strong emphasis on enhancing customer care, which is crucial for clinics. Our software is designed to help small clinics and those with existing offline systems transition to an online platform without compromising service quality. By improving appointment management and optimizing operational processes, MedPal ensures that clinics can focus on delivering the best possible care to patients. This enhances the customer experience, boosts satisfaction, and fosters strong, sustainable relationships between clinics and their patients.

Electronic health record integration streamlines information retrieval, provides instant access to patient history and test results, reduces redundant testing, and promotes decision making insight. Additionally, our patient flow management system optimises clinic operations, reducing wait times and improving scheduling efficiency.

MedPal provides an important business opportunity for small and medium-sized clinics (that already have a management system in place, just don't have a patient care system) to improve services and maintain competitiveness. Looking ahead, we aim to continue upgrading and improving our system to ensure convenience and better support for both clinics and patients.

4.2 Integrated System

MedPal is a solution specifically designed to integrate effectively into the existing systems of small and medium-sized clinics without altering established workflows. Understanding that many clinics have invested in their own management systems and do not want to change well-functioning processes, MedPal was developed as a flexible bridge, ensuring synchronization and expanding the capabilities of the existing system. One of the reasons for integrating MedPal into an existing system is its ability to extract data from captured images or scanned documents, supporting both English and Vietnamese. The Optical Character Recognition (OCR) technology that MedPal utilizes offers many advantages over traditional data entry methods. OCR can process data quickly and automatically, saving time and increasing work efficiency. Instead of manually entering each line, OCR can extract information from documents in seconds, which is especially useful when handling large volumes of data. Additionally, OCR ensures higher consistency and accuracy, reducing human errors.

Instead of having to enter each line of characters and then press a key combination to get data from the input text, by simply capturing or scanning documents such as medical records, prescriptions, or any other medical documents, MedPal automatically recognizes and extracts key information using OCR technology, then synchronizes the data into the existing system without needing to change the main management system. MedPal does not require clinics to overhaul their entire process or current system but simply provides an intelligent integration layer. In this way, small and medium-sized clinics can fully leverage modern technology while maintaining their familiar operational processes, minimizing disruptions.

In summary, with the ability to automate the data entry process through OCR technology integrated into MedPal, clinics not only maintain stability in their operations but also significantly reduce the workload for receptionists. Instead of manually processing each document, receptionists can focus on more important tasks, contributing to increased efficiency and improved service quality at the clinic.

5. Software Product Vision

Recognizing the limitations of the current system, our system offers a solution to overcome these persistent shortcomings. Medpal: A system to improve the quality of customer care services for **68A polyclinic** is a system that provides better solutions for customer care services:

- We provide a method of medical examination and treatment that is easy to

access to customers by creating an online examination registration platform.

- We provide an online search method that customers can look up at home. Our software also provides an online platform for consulting and medical examination.
- Customers can search for information about the clinic through the online platform that we provide to understand the services of the clinic as well as what the clinic has achieved.
- Not only for **68A polyclinic, MedPal** also targets small and medium sized clinics that already have an internal system to improve customer care services.

6. Project Scope & Limitations

6.1 Major Features

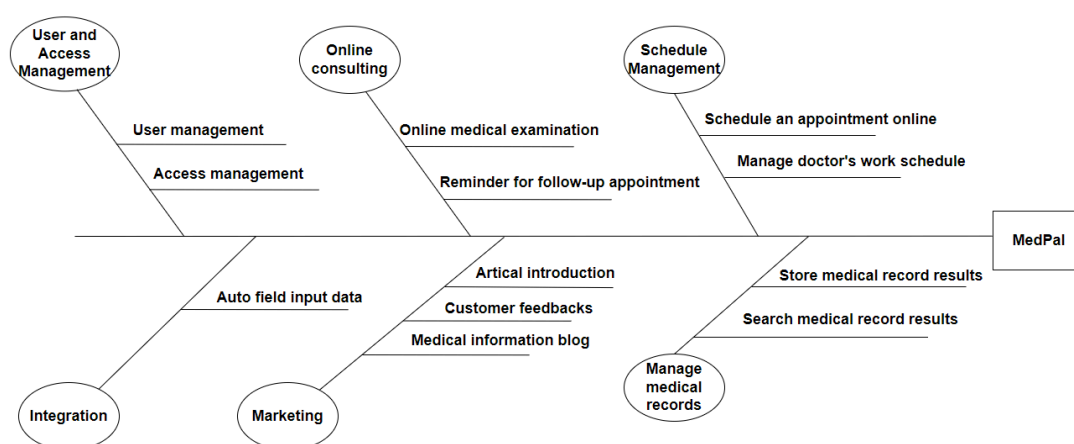


Figure 6.1.1 Major Features tree

FE-01: Schedule an appointment online(Including book, cancel and reschedule)

FE-02: Manage doctor's work schedule (Including view , modify and delete)

FE-03: Store medical record results(including create, view, modify)

FE-04: Search medical record results(customer search medical record result)

FE-05: Brand promotion(Including create , view, modify and delete)

FE-06: Customer feedbacks(Including create, view, modify feedbacks)

FE-07: Medical information blog(including view disease, health care information)

FE-08: Online medical examination(chat with patient)

FE-09: Reminder for follow-up appointment(send notification through SMS, email)

FE-10: Auto field input data from existing system

6.2 Limitations & Exclusions

LI-1: In the medical notebook function, the receptionist needs to hand-capture test results (such as x-rays, general test results, etc.).

LI-2: While a website can provide many conveniences for patients and staff, it cannot entirely replace personal interaction and communication.

LI-3: The doctor does not prescribe on paper but on a computer, which is really

inconvenient for doctors with poor eyesight.

EX-1: The system does not support mobile applications. That means there is no application.

7. Future work

1: Develop a phone app to facilitate monitoring for patients and doctors, instead of having to access the web.

2: Create an integration so the receptionist no longer has to take pictures and update them to the patient's medical notebook manually